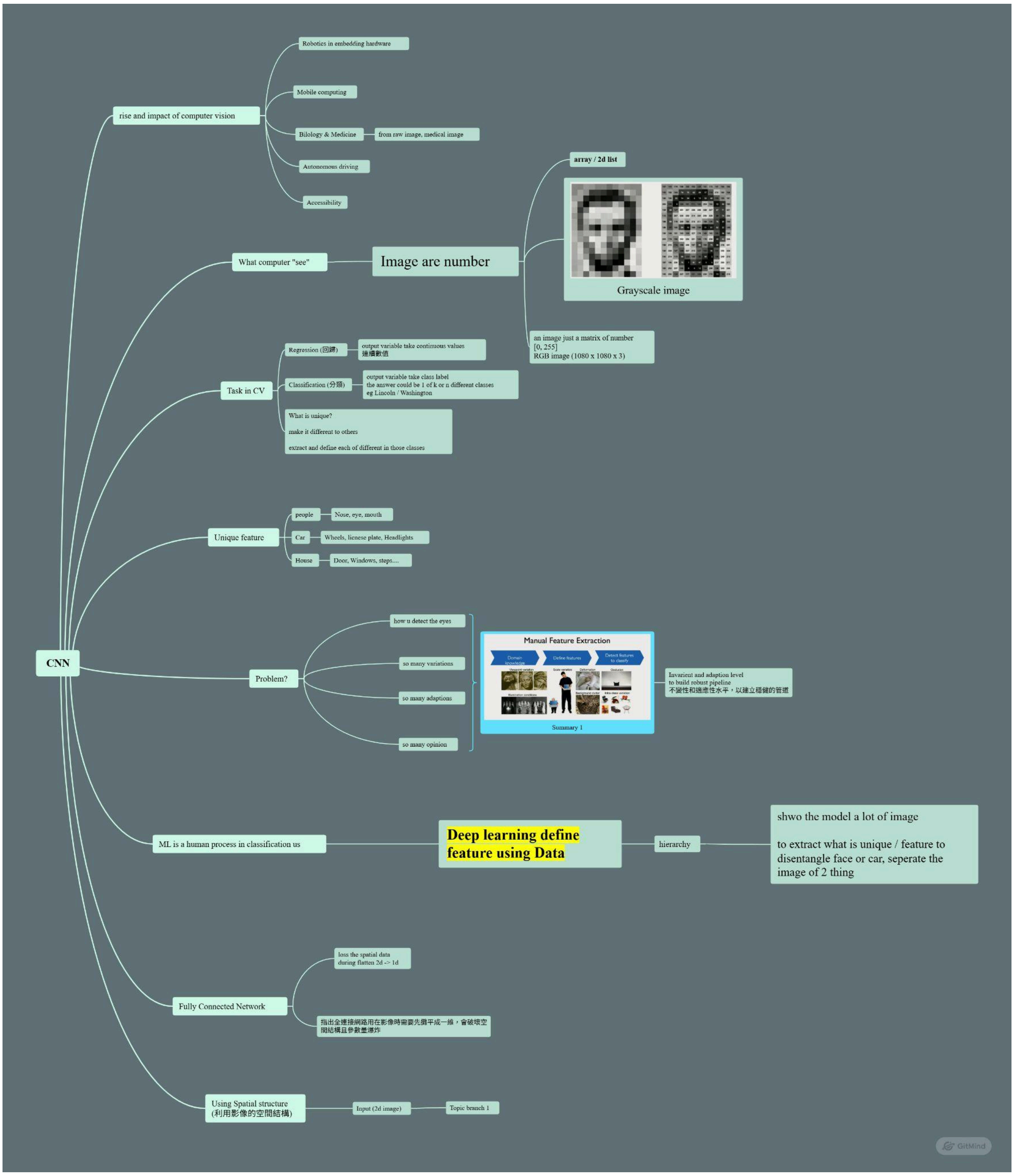
Reflection 2: Introduction to Deep Learning

https://www.youtube.com/watch?v=II4giR4vOOo



During the development included watching an introductory deep learning lecture and taking visual notes summarising the main points. This experience taught me that deep learning teaches a model useful features directly from data, rather than through the process of rule-based manual designing or handcrafted feature engineering. I also realized that images are just numerical arrays; thus, computer vision models need to learn patterns from pixel values.

It helped me understand the difference between traditional manual feature extraction and deep learning. Traditional approaches require human decisions on the important features, like the presence of eyes, edges, textures, or shapes of objects. Deep learning models can automatically learn these features in many layers. The early layers of deep learning models inform the optimal parameters for identifying basic patterns, while deeper layers are involved in identifying much more sophisticated patterns and abstract features. This helped me understand why deep learning powered agreed to work on image classification tasks.

It changed the way I think about my deepfake detection project. I realised it was not just classifying whether an image was real or faked; the model had also learnt hidden visual patterns from the data. Hence, I need to pay attention to the dataset quality, preprocessing, training strategy, and evaluation. In my report, I was able to clearly explain why a CNN/ResNet50-based approach was used because deep learning can automatically learn visual artefacts that may separate real and AI-generated images.